Incidence of dementia and Alzheimer disease in 2 communities: Yoruba residing in Ibadan, Nigeria, and African Americans residing in Indianapolis, I...

CONTEXT: Alzheimer disease (AD) represents a major and increasing public health problem. If populations were identified with significantly lower or higher incidence rates of AD, the search for risk factors in the genesis of AD could be greatly enhanced. OBJECTIVE: To compare incidence rates of dementia and AD in 2 diverse, elderly community-dwelling populations. DESIGN: The Indianapolis-Ibadan Dementia Project, a longitudinal, prospective population-based study consisting of a baseline survey (1992-1993) and 2 subsequent follow-up waves after 2 years (1994-1995) and 5 years (1997-1998). Each wave followed a 2-stage design, with an in-home screening interview followed by a full diagnostic workup of a subsample of participants based on screening performance. SETTING AND PARTICIPANTS: A total of 2459 community-dwelling Yoruba residents of Ibadan, Nigeria, without dementia, and 2147 community-dwelling African American residents of Indianapolis, Ind, without dementia (all aged 65 years or older). The cohorts were followed up for a mean of 5.1 years and 4.7 years, respectively. MAIN OUTCOME MEASURES: Incident cases of dementia and AD in each of the 2 populations. RESULTS: The age-standardized annual incidence rates were significantly lower among Yoruba than among African Americans for dementia (Yoruba, 1.35% [95% confidence interval [CI], 1.13%-1.56%]; African Americans, 3.24% [95% CI, 2.11%-4.38%]) and for AD (Yoruba, 1.15% [95% CI, 0.96%-1.35%]; African Americans, 2.52% [95% CI, 1.40%-3.64%]). CONCLUSION: This is the first report of incidence rate differences for dementia and AD in studies of 2 populations from nonindustrialized and industrialized countries using identical methods and the same group of investigators in both sites. Further explorations of these population differences may identify potentially modifiable environmental or genetic factors to account for site differences in dementia and AD.